

Ensembles

Why the average of many mediocre answers beats one good answer

Carlos Cotrini

Foundations of AI, UZH CAS Gamechanger AI, 26 September 2026

The guessing round

One page, one row, one guess. Write your guess down before you read anybody else's.

① The page

Nothing to type

It picks your code



The page

[https://carloscotrini.github.io/
fai-hs26/marble-jar-guessing/](https://carloscotrini.github.io/fai-hs26/marble-jar-guessing/)

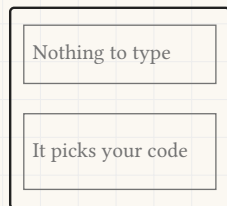
The sheet

[https://placeholder.invalid/
marble-jar-shared-sheet](https://placeholder.invalid/marble-jar-shared-sheet)

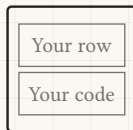
The guessing round

One page, one row, one guess. Write your guess down before you read anybody else's.

① The page



② What you get



The page

[https://carloscotrini.github.io/
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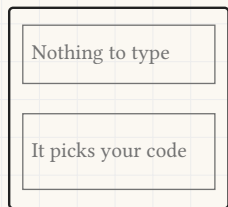
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marble-jar-shared-sheet](https://placeholder.invalid/marble-jar-shared-sheet)

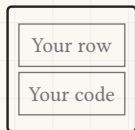
The guessing round

One page, one row, one guess. Write your guess down before you read anybody else's.

① The page



② What you get



③ The sheet

Code	Guess



The page

[https://carloscotrini.github.io/
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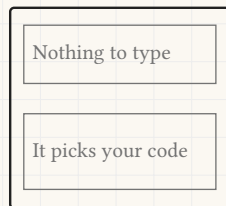
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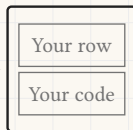
One page, one row, one guess. Write your guess down before you read anybody else's.

① The page



If your row already holds a different code, take the next empty row down.

② What you get



③ The sheet

Code	Guess



The page

<https://carloscotrini.github.io/fai-hs26/marble-jar-guessing/>

The sheet

<https://placeholder.invalid/marble-jar-shared-sheet>

What makes a prediction wrong

One case, and one dot for every dataset we could have been handed. Schematic, not measured.

The error of a prediction

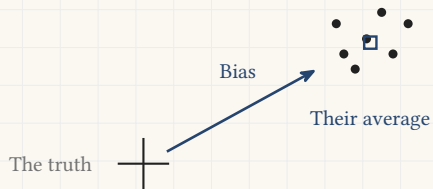


The truth



What makes a prediction wrong

One case, and one dot for every dataset we could have been handed. Schematic, not measured.



The error of a prediction

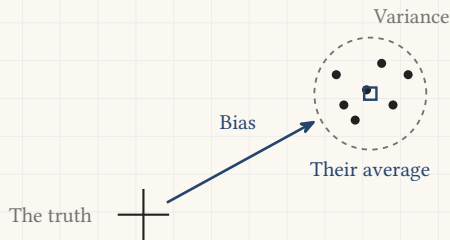
equals

Bias, squared

How far off we are on average, over every dataset we could have been handed

What makes a prediction wrong

One case, and one dot for every dataset we could have been handed. Schematic, not measured.



The error of a prediction

equals

Bias, squared

How far off we are on average, over every dataset we could have been handed

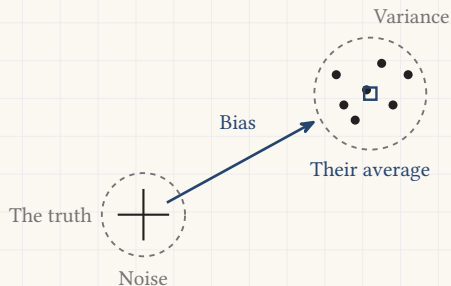
plus

Variance

How much the prediction changes from one dataset to the next

What makes a prediction wrong

One case, and one dot for every dataset we could have been handed. Schematic, not measured.



The error of a prediction

equals

Bias, squared

How far off we are on average, over every dataset we could have been handed

plus

Variance

How much the prediction changes from one dataset to the next

plus

Noise

What nobody could have predicted

What makes a total wobble

Thirty five guesses, added up. Every cell of the table is one piece of the total's wobble.

The guesses

	1	2	3	4
1	●			
2		●		
3			●	
4				●

- How much one guess wobbles on its own

What makes a total wobble

Thirty five guesses, added up. Every cell of the table is one piece of the total's wobble.

The guesses

	1	2	3	4
1	●	○	○	○
2	○	●	○	○
3	○	○	●	○
4	○	○	○	●

- How much one guess wobbles on its own
- How much a pair of guesses wobbles together

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Thirty five guesses, added up. Every cell of the table is one piece of the total's wobble.

The guesses

	1	2	3	4
1	●	○	○	○
2	○	●	○	○
3	○	○	●	○
4	○	○	○	●

- How much one guess wobbles on its own
- How much a pair of guesses wobbles together

- 1 How much a total wobbles is the wobble of each part, plus how much every pair of parts wobbles together.

What makes a total wobble

Thirty five guesses, added up. Every cell of the table is one piece of the total's wobble.

The guesses

	1	2	3	4
1	●			
2		●		
3			●	
4				●

- How much one guess wobbles on its own
- How much a pair of guesses wobbles together

Guesses that do not move together leave every pair cell empty.

- 1 How much a total wobbles is the wobble of each part, plus how much every pair of parts wobbles together.
- 2 Thirty five guesses that do not move together: their average wobbles thirty five times less than one guess does.

What makes a total wobble

Thirty five guesses, added up. Every cell of the table is one piece of the total's wobble.

The guesses

	1	2	3	4
1	●	○	○	○
2	○	●	○	○
3	○	○	●	○
4	○	○	○	●

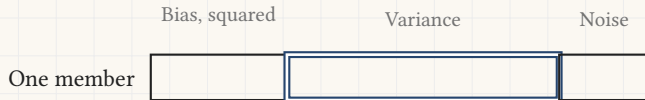
- How much one guess wobbles on its own
- How much a pair of guesses wobbles together

Guesses that move together fill every pair cell, and there are far more pair cells than guesses.

- 1 How much a total wobbles is the wobble of each part, plus how much every pair of parts wobbles together.
- 2 Thirty five guesses that do not move together: their average wobbles thirty five times less than one guess does.
- 3 Guesses that move together keep their wobble, however many of them you add up. The crowd is then one guess wearing thirty five hats.

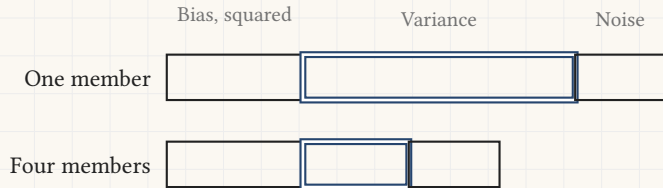
What averaging changes

The same three pieces, for one member and then for many. Schematic, not measured.



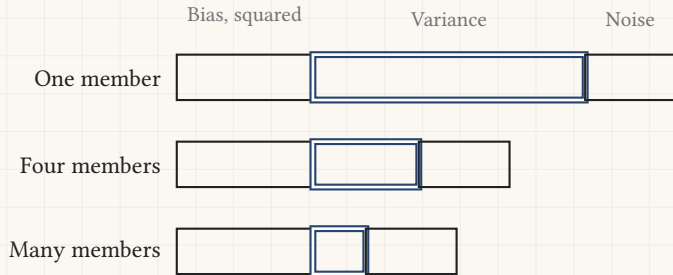
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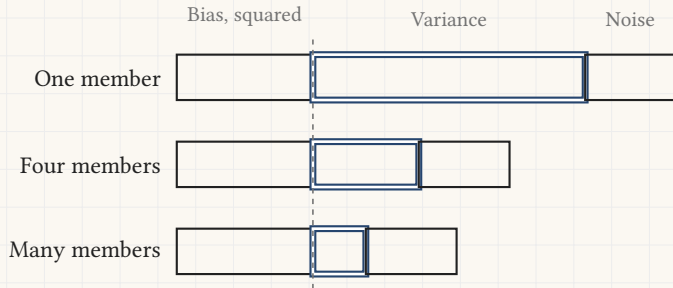
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What averaging changes

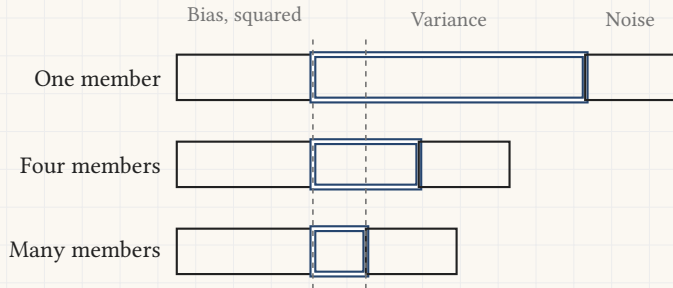
The same three pieces, for one member and then for many. Schematic, not measured.



The bias does not move.
Averaging cannot make a
wrong idea right.

What averaging changes

The same three pieces, for one member and then for many. Schematic, not measured.



The bias does not move.
Averaging cannot make a
wrong idea right.

What is left is set by how much the members
agree with each other, so they have to be built to
disagree.

The forest grows

Eight scenes on two spirals that no single tree can draw.



[https://sml-fs26.github.io/
pca-kmeans-rf/random-forest-deepdive/](https://sml-fs26.github.io/pca-kmeans-rf/random-forest-deepdive/)

The forest grows

Eight scenes on two spirals that no single tree can draw.

- 1 Two spirals
- 2 A tree can only cut one way
- 3 The same data, jittered, gives a different tree
- 4 Sample with replacement



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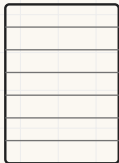
- 1 Two spirals
- 2 A tree can only cut one way
- 3 The same data, jittered, gives a different tree
- 4 Sample with replacement
- 5 Sixteen trees, sixteen staircases
- 6 Average the votes
- 7 How many trees is enough
- 8 The diversity ladder



[https://sml-fs26.github.io/
pca-kmeans-rf/random-forest-deepdive/](https://sml-fs26.github.io/pca-kmeans-rf/random-forest-deepdive/)

How a random forest is grown

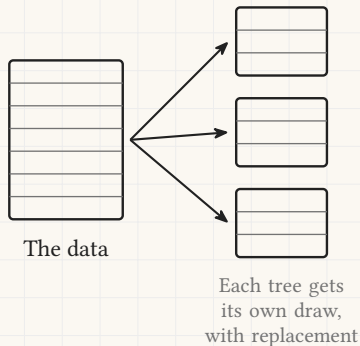
The recipe, and the two places a die is rolled.



The data

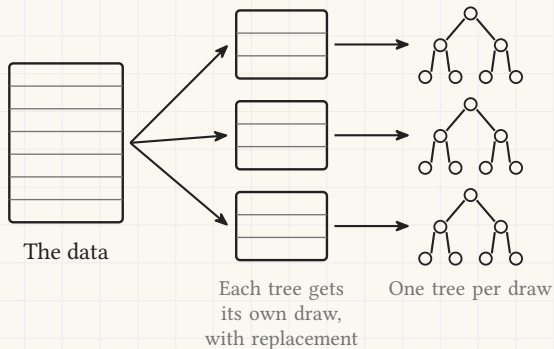
How a random forest is grown

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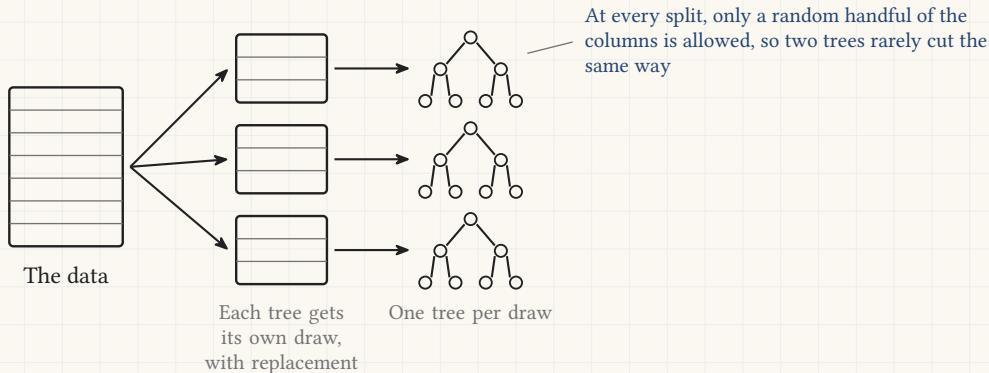
How a random forest is grown

The recipe, and the two places a die is rolled.



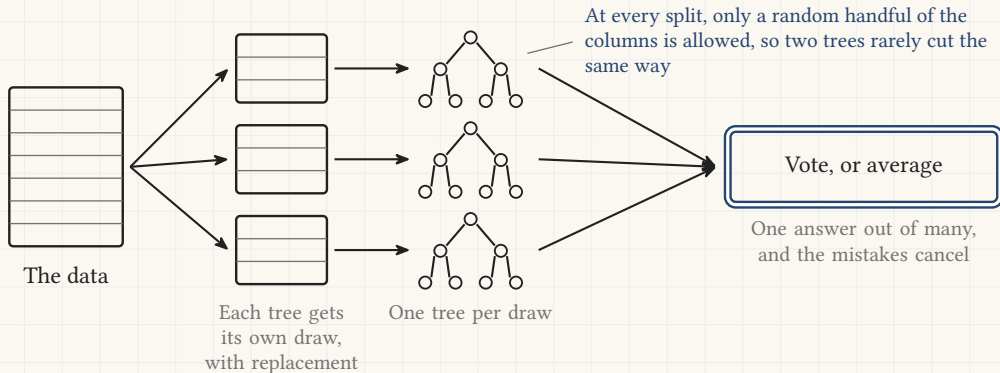
How a random forest is grown

The recipe, and the two places a die is rolled.



How a random forest is grown

The recipe, and the two places a die is rolled.



The UN games

Eleven scenes in five chapters, on a model that scores well and knows nothing.



[https://carloscotrini.github.io/
FAI-PAGE/viz/un-games/](https://carloscotrini.github.io/FAI-PAGE/viz/un-games/)

The UN games

Eleven scenes in five chapters, on a model that scores well and knows nothing.

- 1 The bait: one feature at a time, then all 493 at once
- 2 The crash: fit everything, then only nonsense



[https://carloscotrini.github.io/
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The UN games

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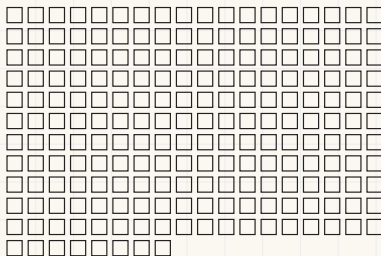
- 1 The bait: one feature at a time, then all 493 at once
- 2 The crash: fit everything, then only nonsense
- 3 The patch: regularisation, and the score comes back
- 4 The reveal: two models that predict equally well disagree about what mattered
- 5 The trust: one split, and then a hundred



[https://carloscotrini.github.io/
FAI-PAGE/viz/un-games/](https://carloscotrini.github.io/FAI-PAGE/viz/un-games/)

The envelope

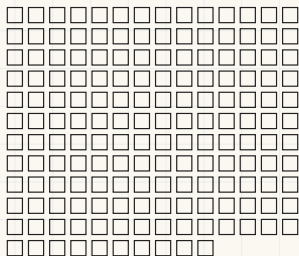
Before the model sees anything, the countries are dealt into two piles.



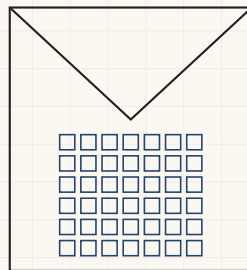
206 countries

The envelope

Before the model sees anything, the countries are dealt into two piles.



The desk: 164 countries (train)

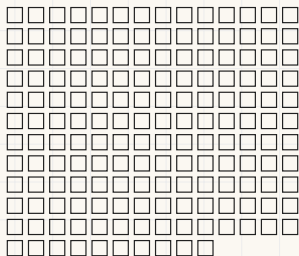


The envelope: 42 countries (test)

The envelope

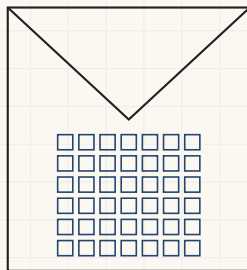
Before the model sees anything, the countries are dealt into two piles.

The model learns from these



The desk: 164 countries (train)

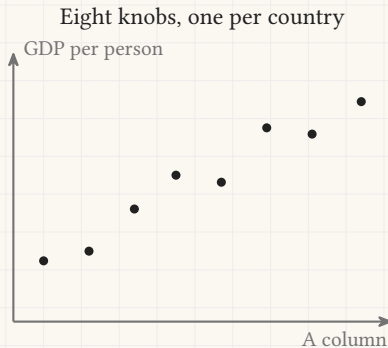
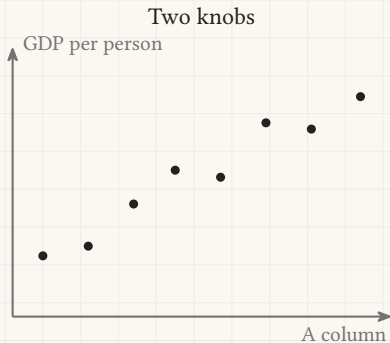
Its score is read on these



The envelope: 42 countries (test)

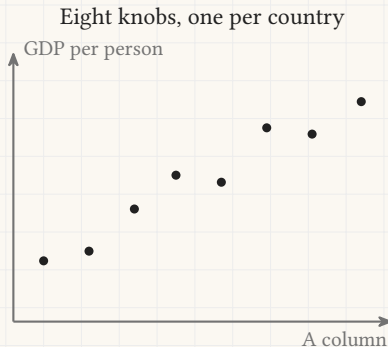
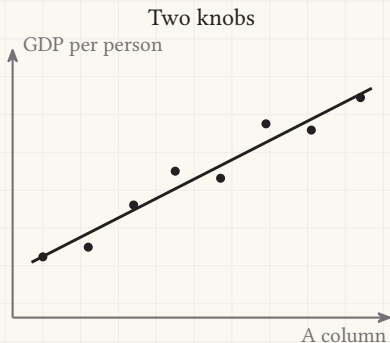
164 countries, 164 knobs

One knob per country, and the curve can pass through every country it saw. Schematic, not measured.



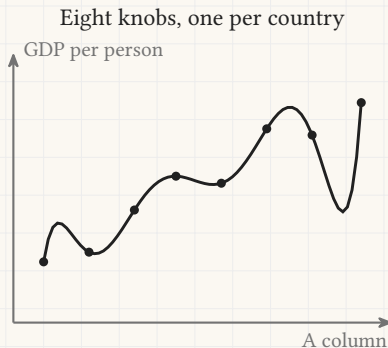
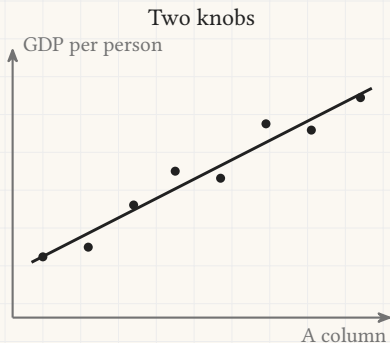
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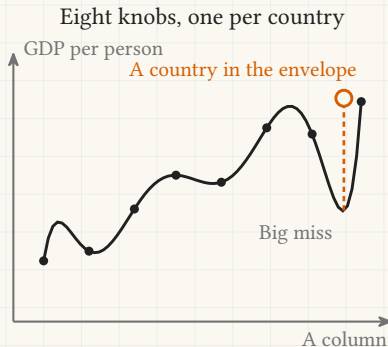
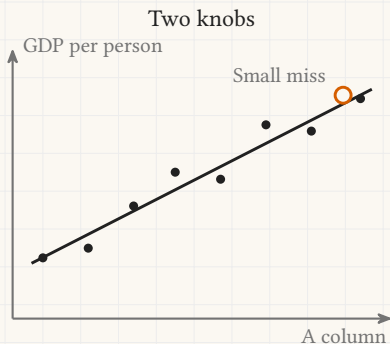
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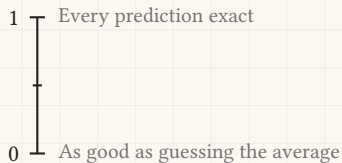
The score: R squared

The score compares the model's misses with the misses of guessing the average for every country.

1 — Every prediction exact

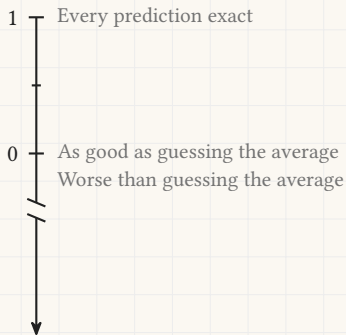
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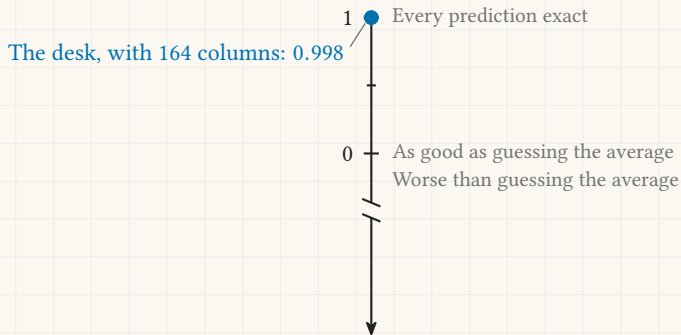
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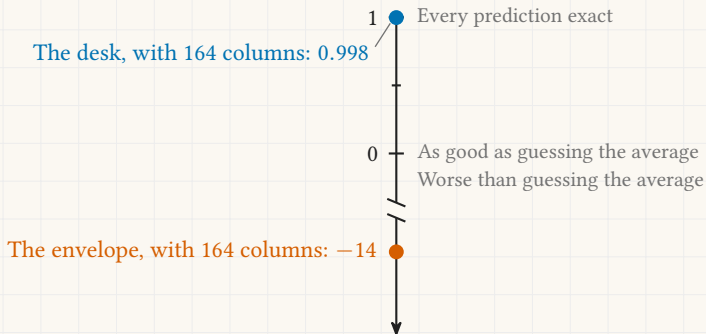
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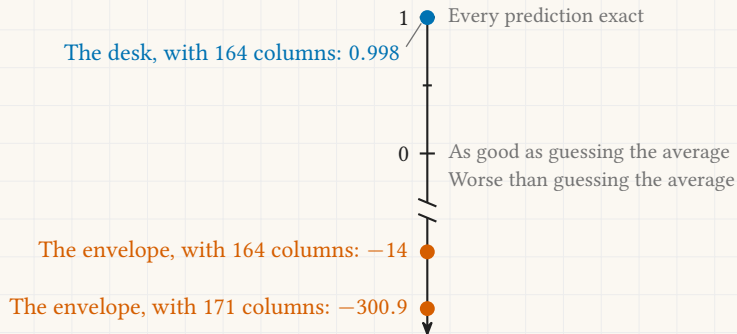
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The score: R squared

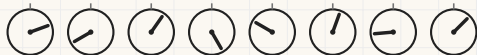
The score compares the model's misses with the misses of guessing the average for every country.



A tax on knobs

Each knob now pays a fee for how far it turns.

No tax

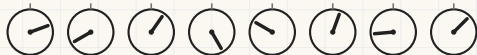


Least squares
Envelope: -14

A tax on knobs

Each knob now pays a fee for how far it turns.

No tax



Least squares

Envelope: -14

Ridge

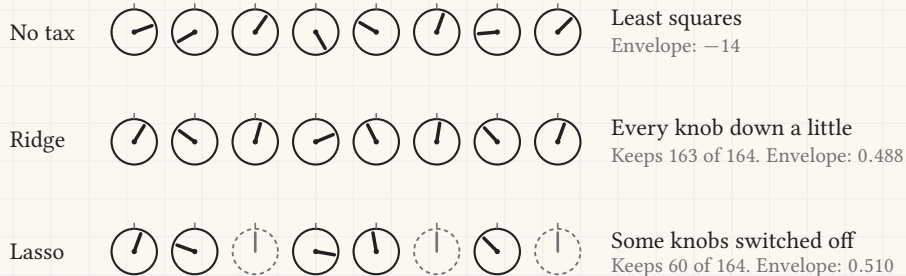


Every knob down a little

Keeps 163 of 164. Envelope: 0.488

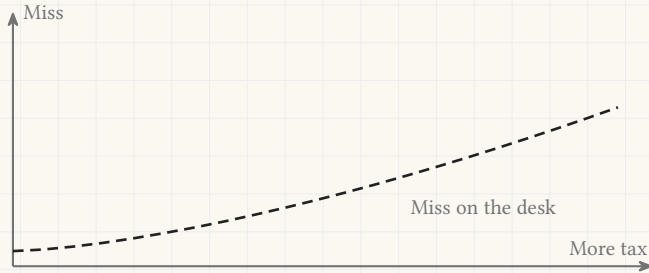
A tax on knobs

Each knob now pays a fee for how far it turns.



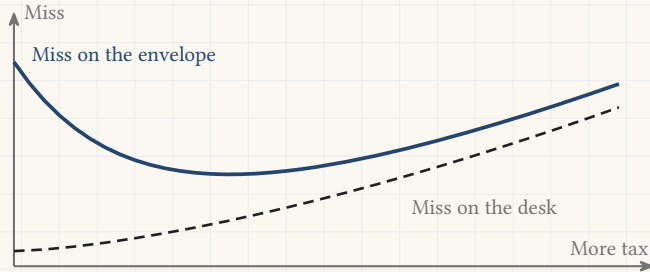
Goldilocks

Too little tax, too much tax, and in between. Schematic, not measured.



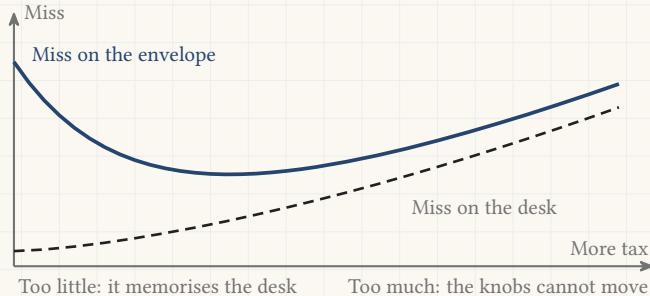
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